

BUSINESS INTELLIGENCE AND TOOLS

(24CSH-341)

EXPERIMENT 2 : End-to-End ETL and Sales Analysis in Tableau

AIM To design and implement a complete ETL (Extract, Transform, Load) pipeline using the Tableau ecosystem (Tableau Prep Builder and Tableau Desktop) to extract data from a real-world business dataset, apply necessary data transformation logic, and load the clean data into a unified analytical model for reporting and decision-making.

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OBJECTIVES

- To understand the ETL process and its crucial role in Business Intelligence.
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 - To extract raw data from the Global Superstore dataset.
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 - To clean, transform, standardize, and integrate this data using Tableau Prep Builder (the equivalent of Power Query Editor).
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 - To establish logical relationships between multiple tables in the Tableau Data Model.
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 - To create analytical measures using Tableau Calculated Fields (replacing DAX).
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 - To prepare the newly transformed data for comprehensive reporting and business analysis.
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TOOLS & SOFTWARE REQUIRED

- Tableau Prep Builder (for the Extract and Transform phases)
- Tableau Desktop (for the Load and Visualization phases)
- **Dataset Used:** [Global Superstore Dataset.xlsx](#)

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THEORY

The ETL Pipeline in Tableau ETL stands for Extract, Transform, and Load—a foundational process in Business Intelligence used to gather data from source systems, format it cleanly, and centralize it for analysis.

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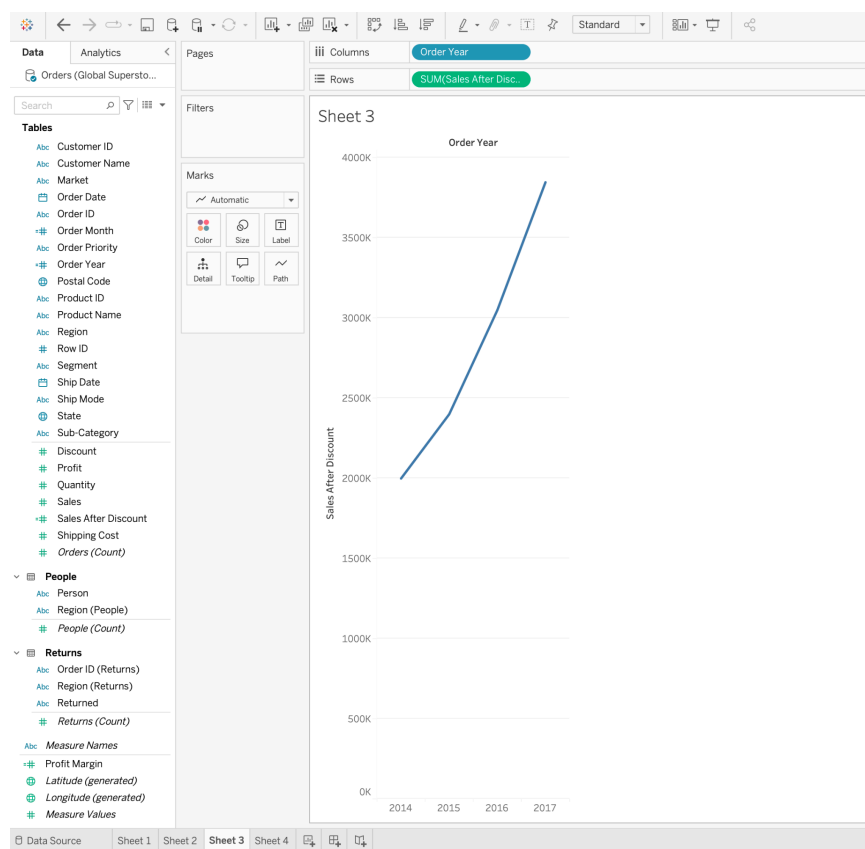
- **Extract:** This is the gathering phase. For this experiment, we extract data from three specific worksheets within the Global Superstore dataset: Orders, Returns, and People.
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 - **Transform:** This is the cleaning phase. Raw data is rarely perfect. Transformation involves removing duplicates, deleting blank rows, correcting data types, renaming columns, deriving new columns (like Profit Margin), and standardizing values to ensure consistency. In the Tableau ecosystem, Tableau Prep Builder acts as the primary ETL engine, recording every transformation as a step in a visual flow so the process is transparent and reproducible.
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 - **Load:** The fully transformed data is finally loaded into Tableau Desktop's data model. Here, we can build relationships between tables and create analytical measures to power our dashboards.
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PROCEDURE

Part A: Extract – Connect to the Dataset

- **Step 1: Download the Dataset**

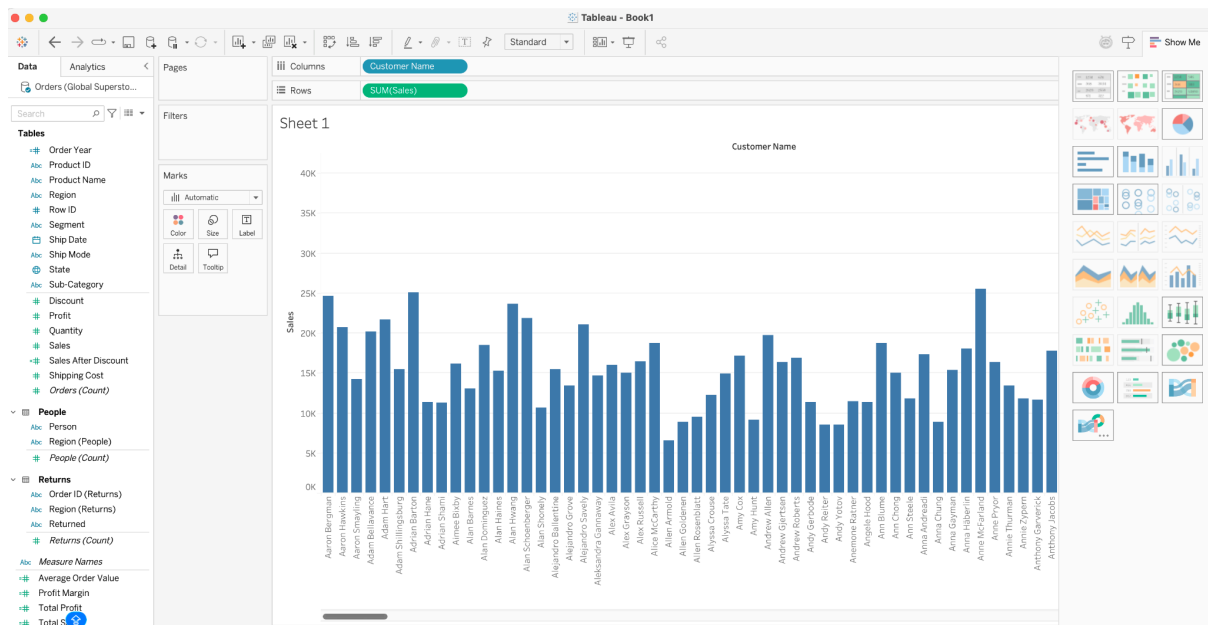
- Secure the **Global Superstore Dataset.xlsx** file and save it in your working directory.
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- **Step 2: Connect via Tableau Prep Builder**
 - Open Tableau Prep Builder.
 - Click **Connect to Data > Microsoft Excel**, and select **Global Superstore Dataset.xlsx**.
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- **Step 3: Import the Worksheets**
 - Drag the **Orders**, **Returns**, and **People** worksheets onto the flow canvas.
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 - Verify that all three tables represent distinct input steps in your workflow.
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Part B: Transform – Cleaning the Data (Tableau Prep Builder)

- **Step 4: Clean the Orders Table**

- Add a **Clean Step** to the Orders input. Apply the following transformations:
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 - Remove any duplicate records.
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 - Filter out or remove completely blank rows.
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 - Check the profile pane to verify and correct data types (e.g., ensuring ZIP codes are strings, not measures).
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 - Rename columns for clarity where required.
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 - Use a calculated field to extract the **Year** and **Month** from the Order Date.
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 - Create a derived column for **Profit Margin**.
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 - Group and replace values to standardize the Region and Segment fields.
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- **Step 5: Clean the Returns Table**
 - Add a Clean Step to the Returns input.
 - Remove duplicate Order IDs.
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 - Check the data grid for any missing values.
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 - Standardize the text in the "Returned" field (e.g., ensuring all values read as simply "Yes").
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- **Step 6: Clean the People Table**
 - Add a Clean Step to the People input.
 - Validate the regional assignments for each salesperson.
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 - Remove any unnecessary columns that will not be used in analysis.
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 - Ensure the region names match exactly with the regions in the Orders table.
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- **Step 7: Merge (Join) the Datasets**
 - Create a **Join Step** by dragging the Cleaned Returns table onto the Cleaned Orders table.
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 - Configure the join condition using the **Order ID** field.
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- **Step 8: Output the Transformed Data**
 - Add an **Output Step** at the end of your flow.
 - Run the flow to generate a **.hyper** extract file (Tableau's optimized data format).

Part C: Load – Build the Data Model and Visuals (Tableau Desktop)

- **Step 9: Open Tableau Desktop & Create Relationships**
 - Open Tableau Desktop and connect to your newly created **.hyper** extract.
 - If bringing tables in separately (instead of joining in Prep), use the logical canvas to relate the tables:
 - Relate **Orders** to **Returns** using Order ID.
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 - Relate **Orders** to **People** using Region.
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 - Verify that the relationships (noodles) are established.
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- **Step 10: Create Analytical Measures (Calculated Fields)**
 - Right-click in the Data pane and select **Create Calculated Field** for the following:
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 - **Total Sales:** `SUM([Sales])`

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- **Total Profit:** $SUM([Profit])$

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- **Profit Margin:** $SUM([Profit]) / SUM([Sales])$

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- **Total Orders:** $COUNTD([Order\ ID])$ (Tableau's equivalent to DISTINCTCOUNT)

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- **Step 11: Build the BI Dashboard**

- Open a new Dashboard canvas and create the following visualizations:

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- **A Sales by Category** bar chart.

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- **A Profit by Category** pie chart.

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- **A Profit Margin vs Discount** scatter chart.

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- **A Detailed Sales & Profit Analysis** text table.

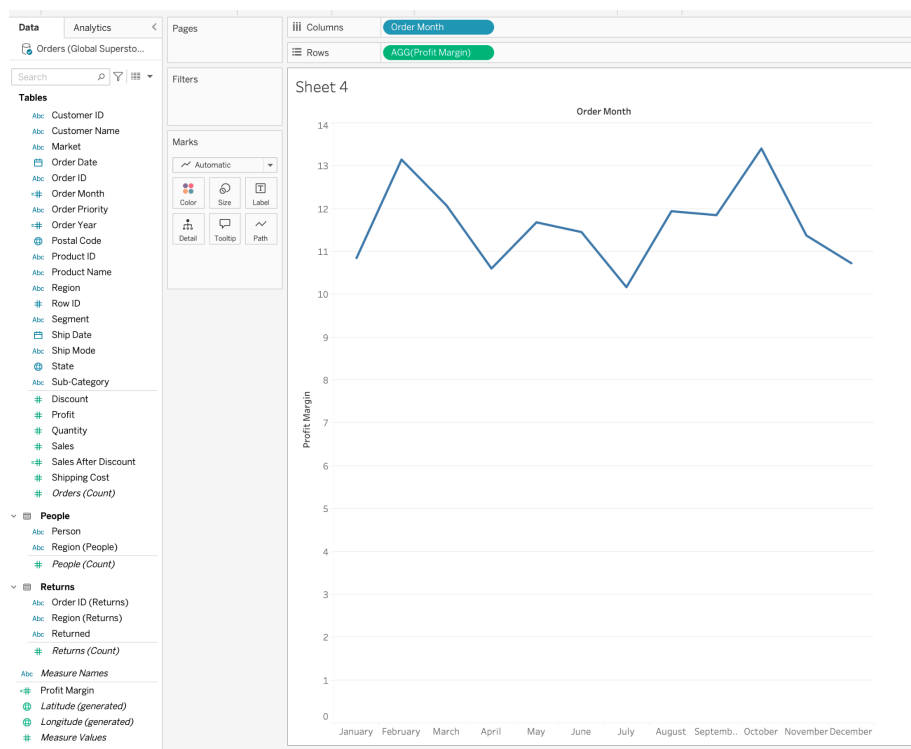
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- Include KPI text boxes at the top for Total Sales, Total Profit, Profit Margin, and Total Orders.

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- Add a prominent text object for the dashboard title: **GLOBAL SUPERSTORE – ETL & SALES ANALYSIS.**

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RESULT

Data was successfully extracted from the Global Superstore Excel dataset and run through a robust ETL pipeline. Using Tableau Prep Builder, the data underwent rigorous cleaning, standardization, and joining. The polished dataset was then loaded into Tableau Desktop's data model, where relationships were verified, and key business metrics were generated using Calculated Fields. The final outcome is a comprehensive, interactive dashboard perfectly prepared for business analysis and reporting.

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LEARNING OUTCOMES

- Gained a clear understanding of the ETL process and its essential role in Business Intelligence.
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- Learned how to extract real-world business data effectively.
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- Acquired hands-on practical experience utilizing Tableau Prep Builder for data transformation.
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- Learned how to clean up dirty data by filtering out blanks, fixing data types, and handling duplicate records.
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- Mastered the process of joining and merging datasets.
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- Learned how to establish logical relationships between multiple tables in Tableau.
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- Understood how to translate DAX concepts into Tableau Calculated Fields (Total Sales, Total Profit, Profit Margin, Total Orders).
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- Successfully created a suite of BI visualizations combined into a cohesive dashboard for decision-making.
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